PAKISTAN NAVY · DIRECTORATE OF MEDIA

RESTRICTED · OPERATIONAL USE

Channel Monitoring & Clipping

Software Manual

Broadcast intelligence platform for PAK NAVY operators & analysts

CMC · VERSION 1.0

VERSION	ISSUED	CLASSIFICATION
1.0.0	August 2026	Internal

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CONTENTS

What's in this manual

Everything you need to run CMC end-to-end — from your first sign-in through building a broadcast-quality clip for evidence or briefing. Chapters are grouped into six parts; each chapter opens on a fresh page.

01 **PART I · GETTING STARTED**

02 Introduction p. 4
What CMC does, who uses it, key concepts.

03 Signing in p. 6
URL, credentials, first-launch behaviour.

04 The interface p. 8
Layout of the app, navigation, common conventions.

05 **PART II · MONITORING**

06 Dashboard p. 10
KPI tiles, live health heatmap, recent alerts.

07 Channels p. 12
Browsing the channel list, source health, per-channel detail.

08 The Video Wall p. 15
Installing and running CMC-VideoWall on the operator PC.

09 **PART III · RECORDING & CLIPPING**

10 24×7 Recording p. 19
How recording works, storage layout, retention.

11 Browsing recordings p. 21
Calendar, hourly strip, segment list, in-place playback.

12 Downloading segments p. 23
Single-segment download and multi-segment stitching.

13 Creating clips p. 25
Opening the NLE editor, in/out markers, watermark, export.

14 Managing clips p. 28
Listing, previewing, downloading, deleting.

15 PART IV · DOWNLOADS

16 Downloads panel p. 30
Getting the Video Wall installer and other assets.

17 PART V · ADMINISTRATION

18 Users & roles p. 31
Creating operator accounts, disabling access, roles.

19 Audit log p. 33
Reviewing every state-changing action for accountability.

20 Settings p. 34
Retention days, air-gap mode, system-wide policy.

21 Legal holds p. 35
Locking specific windows from janitor deletion.

22 PART VI · REFERENCE

23 Troubleshooting p. 36
Symptoms, likely causes, first actions to take.

24 Appendices p. 38

Keyboard shortcuts, URL reference, support contact.

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PART I · GETTING STARTED

01 Introduction

What CMC is, who it serves, and the model behind the platform.

What CMC does

Channel Monitoring & Clipping (CMC) ingests every TV feed of interest to the Pakistan Navy — currently thirty DGPR-approved news channels — off the satellite UDP multicast, decodes them centrally, and makes them available three ways:

- **Live wall** — every channel visible at once on the operations-room wall via the *CMC-VideoWall* desktop viewer.
- **Continuous recording** — 24×7 storage of every channel on a rolling 30-day window, indexed to the second, evidence-grade with per-segment SHA-256 checksums.
- **Clip creation** — an NLE-style editor for cutting a specific moment out of a recording, adding a burned-in watermark, and exporting the result as an MP4 for briefings or evidence retention.

Who uses it

CMC is designed for four operator profiles. Each has its own dashboard posture, its own set of permissions, and its own workflow.

ADMIN **System owner**

Runs the platform. Manages users, keeps an eye on the audit log, approves legal holds, changes retention policy. Typically the on-duty NCO for IT.

OPERATOR **Wall operator**

Runs the video wall day-to-day, drops in on live feeds, kicks off quick clip exports for the duty officer. Doesn't touch user accounts or system settings.

ANALYST **Content review**

Spends most time in the Recordings and Clips sections — reviewing past broadcasts, cutting the sequences of interest, exporting for onward distribution.

VIEWER **Read-only**

Can browse recordings and clips but cannot download or export. Assigned to auditors and to observers who need visibility without the ability to move media off the platform.

The model behind the platform

Three concepts recur throughout the interface. Understanding them makes everything else obvious.

CHANNEL
A logical TV network

SOURCE
One physical UDP feed of that channel

SEGMENT
A 60-second slice of recording

A channel like *Geo News* may have one **source** today and a backup source added tomorrow — the channel is the operator's view, sources are how it actually arrives. Recordings are always broken into 60-second **segments** so the janitor can trim old material precisely and clip exports can start on any even minute boundary.

§ **NOTE**

CMC is a stripped-down deployment of the wider DEMP broadcast intelligence platform, tailored for PAK NAVY's brief: monitor, record, clip. AI transcription, ticker-OCR, and automatic source failover are all out of scope for this build.

PART I · GETTING STARTED

02 Signing in

How to reach the platform, sign in, and what happens on first launch.

The address bar

CMC is delivered as a web application. Open any modern browser (Chrome, Edge, Firefox) and navigate to:

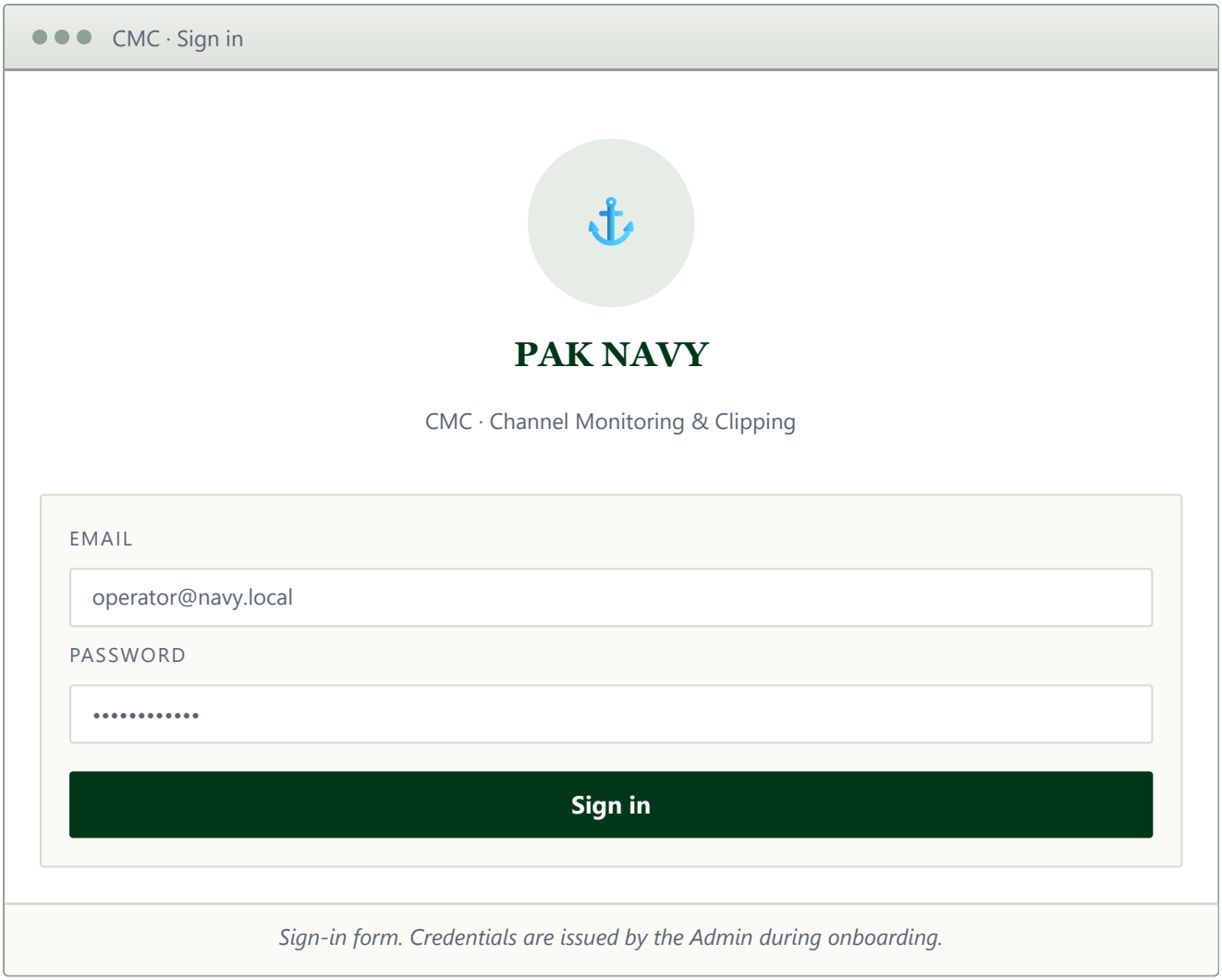
`http://10.51.80.202:3001`

The `10.51.80.202` host is the operations-room server on the recording LAN. If you are on a machine attached to the storage network instead of recording, use `http://172.16.205.21:3001` — both point to the same server.

> ☞ **TIP**
>
> Bookmark the URL on first visit. The dashboard is where every workflow starts and you will return to it often.

The sign-in screen

The sign-in screen carries the PAK NAVY crest at the top of the form. Enter your issued email and password.



Sign-in form. Credentials are issued by the Admin during onboarding.

What happens on first sign-in

1. **The dashboard opens.** A short "welcome" toast in the top-right corner confirms who you are signed in as.

2. **Session persists.** Your browser keeps the session alive for eight hours. Close the tab, come back, no re-login required inside that window.

3. **Role determines what you see.** An Analyst will see the Recordings and Clips sections in the nav; a Viewer will see them but download buttons will be hidden.

! **WRONG CREDENTIALS**

The form shows "Wrong email or password" for a bad login, and "Session expired — please sign in again" when a saved session has timed out. The distinction matters: the first means retry the password; the second usually means you left the tab open overnight and Windows locked the session.

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PART I · GETTING STARTED

The interface

03

A quick tour of the layout so every subsequent chapter has a common vocabulary.

The top bar

Every screen in CMC carries the same top bar. From left to right:

- **PAK NAVY crest & wordmark** — click to return to the dashboard.
- **Section links** — Dashboard, Channels, Recordings, Clips, Downloads, Admin. Admin only appears for accounts with the ADMIN role.
- **Global search** — type a channel name or clip title, jump straight to it.
- **User menu** — your email, role badge, Sign out.

Common patterns

Wherever these appear, they behave the same way.

ELEMENT	BEHAVIOUR
Green primary button	Commits the current action (Sign in, Export clip, Add user). Only one primary button per screen.
Outline secondary button	Non-committing action (Cancel, Reset filters, Back). Always safe to click.
Live dot · ●	Green = channel is delivering packets; grey = channel is disabled or offline.
Duration · 01:42:07	Always *hours:minutes:seconds*. Times in the app are always the server's local time (PKT).
Toast (top-right)	Confirms a save, warns of a failure, links to the affected record. Auto-dismisses in 6 s.

§ **NOTE**

The interface is desktop-first — every screen is built for a 1920 × 1080 monitor. It works on a laptop but the recordings calendar and clip editor really want the horizontal real estate.

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PART II · MONITORING

Dashboard

04

A single glance at how the platform is doing — and where to look if something is off.

Four KPI tiles

The top row summarises the last 24 hours at a glance. Numbers are refreshed live every fifteen seconds — no reload required.

TILE	WHAT IT COUNTS	WHEN TO WORRY
Channels online	Sources delivering packets right now.	Below **28 of 30** during business hours.
Recordings 24 h	New segments written in the last day.	Below **~40,000** across all channels.
Disk usage	Recording volume utilisation.	Above **85%** — call IT before it hits 95.
Clips today	Clips exported since midnight.	Rarely — this is an activity signal, not health.

Live health heatmap

The heatmap below the KPI tiles shows every channel as a coloured square: green = live and healthy, amber = degraded (packet loss), red = down, grey = disabled. Hover any square to see the channel name and its current bitrate; click it to jump straight to the channel detail.

Recent activity strip

Two feeds sit side by side at the bottom of the dashboard:

- **Recent alerts** — the last twenty alerts fired. Currently limited to *source switch* events. Click any row to jump to the affected channel with its state at that moment.
- **Recent clips** — the last twenty clip exports across the platform, with the operator who exported and how long the source was. Click any row to preview the clip.

> ☞ **TIP**
>
> The dashboard is where every duty handover starts. First action of every shift: open the dashboard, check the four tiles are green, glance at the heatmap for any red squares, brief the incoming operator.

PART II · MONITORING

Channels

05

The full channel roster — configure, health-check, drill into any single feed.

The channel list

The Channels page shows every configured channel — logo, name, category, language, live health, current signal, and whether recording is enabled. Sortable and filterable by category or language.

COLUMN	EXPLANATION
Channel	The channel logo (upload from the detail page) alongside its display name and slug.
Category	Editorial category (News, Sports, Entertainment). Set by the Admin.
Lang	Primary language of the feed. Used by the search and by the language filter.
Health	A rolling 60-second bar. Green ticks = healthy; amber = degraded; red = dropped.
Signal	The current one-word verdict — LIVE, DEGRADED, OFFLINE, DISABLED.
Recording	Green pill = recorder is writing segments right now; grey = paused or disabled.

Channel detail

Click any channel row to open its detail drawer. The detail view shows:

Overview

Logo, name, category, language, current signal, and disk usage for this channel over the last day. A small *Upload logo* control at the top lets an Admin drop in a PNG or JPG that will then appear everywhere this channel is shown across the platform.

Recording status

A live badge shows whether the recorder is writing this channel right now, alongside the size of the last hour's segments and the timestamp of the most recent write. If recording has stalled, this is the first place the anomaly shows.

> § **CHANNEL CONFIGURATION IS PRE-PROVISIONED**
> The list of channels and their underlying inputs is pre-provisioned in the backend during platform deployment. Operators, Analysts and Admins view the roster and health through this page but the roster itself doesn't change from inside the web app.

PART II · MONITORING

06 The Video Wall

CMC-VideoWall — the desktop client that drives the physical operations-room wall.

The video wall is the physical display in the operations room — eight monitors in a 4×2 grid, driven from a single Windows workstation. That workstation runs **CMC-VideoWall.exe**, a native desktop client that:

- Signs in against the same server as the web app.
- Joins the satellite UDP multicast for every channel directly, bypassing the server.
- Composes them into one of four fixed layouts.
- Runs on a stripped, portable Rust / GStreamer stack — no install, no admin.

Installing on the wall PC

1. Sign into CMC on any browser. Go to **Downloads → CMC-VideoWall.zip** and download it (≈ 45 MB).

2. Copy the ZIP onto the wall PC. Extract it anywhere on a local drive — `C:\CMC-VideoWall\` is the convention, but the folder is fully relocatable so a USB stick works too.

3. Double-click `CMC-VideoWall.cmd`. The first launch opens the sign-in dialog. Enter your CMC credentials and the API address of the server (`http://10.51.80.202:4001`). The tool remembers you on every subsequent launch.

> ☞ **MULTICAST NIC**
>
> If the wall PC has more than one network card (typical: one for LAN, one for the recording network), fill in the *Multicast NIC IP* field with the interface that sees the satellite feeds — usually `10.51.80.x`. Leaving it blank lets Windows pick, which sometimes picks the wrong card and no channels appear.

The main window

Once signed in, the app shows a preview of the wall in a normal window. From left to right along the top:

- **Layout picker** — dropdown with the four available layouts (Single, Dual, Mosaic-20, Grid-32). Switching layout is instant.
- **Refresh** — re-pull the channel list from the server (use this after Admin adds a new channel).
- **Auto-fill** — checkbox. When on, any empty slot on the wall is filled by cycling through unassigned channels. Turn off if you want empty slots to stay empty.
- **Logout** — sign out and return to the login dialog.
- **Start Fullscreen** — the big green button. Opens the wall as a borderless fullscreen window filling the whole desktop (which the wall's video splitter feeds to the eight physical monitors).

The right-hand panel is the *channel picker*: every channel with its logo, name, and a green / grey dot for online status. The centre is the *wall preview*: coloured tiles laid out in the current layout, each showing the channel logo and name (or "Slot N" when empty).

Placing channels on the wall

1. Click a channel on the right-hand panel — it highlights.
2. Click a slot on the preview — the channel binds there and its logo appears in that slot.

3. Repeat for other channels / slots. Assignments save automatically to your local profile.

4. Right-click any slot for a menu: *Fill mode* (Stretch / Crop-fill / Fit) and *Clear slot*.

Fullscreen playback

Press `F11` or click **Start Fullscreen**. A borderless window opens covering the entire desktop, and every assigned slot starts playing its channel. Empty slots stay black unless *Auto-fill* is on, in which case they're populated automatically.

Press `Esc` or `F11` again in the fullscreen window to return to the preview.

> ! **GRID-32 SIZING**
> Grid-32 puts 32 concurrent 1080p decoders on the wall PC. On the operations-room workstation (10th-gen Intel with iGPU) this runs comfortably. On a laptop or via RDP the CPU may saturate — stick to Mosaic (20 tiles) for those environments.

PART III · RECORDING & CLIPPING

07 24×7 Recording

How recording actually works underneath, so you know exactly what's on disk.

What gets recorded

Every channel with *Recording enabled* in its configuration is recorded continuously — 24 hours a day, 7 days a week, regardless of whether anyone is looking. The recorder runs as a server-side service; nothing on the operator PC is involved.

How it's stored

The recorder chops each channel's live stream into **60-second MP4 segments**. Each segment is written to disk as soon as its 60 seconds are over, then hashed with SHA-256 and indexed in the database.

On disk, segments live under a strict path structure so a human can navigate them without the app:

`/data/recordings/{channelId}/{yyyy}/{mm}/{dd}/{hh}/{mm}.mp4`

Alongside each `.mp4` sits a `.sha256` sidecar with the checksum, and once an hour the segments are grouped into a signed *manifest* — this is the chain-of-custody artefact evidence reviewers ask for.

Retention

A **janitor** service runs every night at 02:00 PKT. It deletes any segment older than the configured retention window (default: **30 days**) UNLESS a legal hold covers that channel's time window — see Legal holds.

METRIC	VALUE	NOTES
Retention window	30 days	Fixed policy — cannot be increased or decreased from the app.
Segment length	60 seconds	Enforced platform-wide.
Storage bitrate	~2 Mbps per channel	Source dependent (satellite feed).
Daily volume	~21 GB / channel / day	Derived from the above.
Storage per channel · 30 d	~630 GB	Sizing target the platform is provisioned for.

§ **RETENTION IS FIXED AT 30 DAYS**

The 30-day rolling window is a platform-wide policy. It is not exposed as an operator-editable setting because storage sizing is calibrated to exactly that window. To keep material beyond 30 days, place a *legal hold* (Chapter 16) or export a clip (Chapter 10).

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PART III · RECORDING & CLIPPING

08 Browsing recordings

Finding the specific broadcast you need — from a month's worth of tape down to the second.

Three-step drill-down

The Recordings page is designed to be drilled in three clicks: channel → day → hour → the specific minute you care about.

1. **Pick a channel.** A dropdown at the top of the Recordings page lists every channel with recording enabled. Type-ahead search makes the pick fast even with thirty channels.

2. **Pick a day.** A month calendar appears. Days with recorded content are highlighted; days with no recording (channel was down, or beyond retention) are greyed out. Click any highlighted day.

3. **Pick an hour.** A horizontal 24-hour strip appears below the calendar, each hour shown as a bar. The bar's height indicates how many minutes in that hour actually recorded (60 = a perfect hour, less = there were gaps). Hover to see a thumbnail; click to open that hour's segment list.

4. **Play a segment.** The segment list is a scrollable table of every 60-second segment in that hour. Click one and an inline player opens showing that segment with a scrubber. Use the transport controls or the keyboard shortcuts (`J` back, `K` pause, `L` forward, `Space` toggle).

Segment metadata

Every row in the segment list shows:

- **Time** — the start-of-segment in HH:MM:SS (PKT).

- **Duration** — normally 60 s, occasionally shorter if the source dropped mid-segment.
- **Size** — how many megabytes the segment occupies on disk.
- **SHA-256** — evidence-chain checksum. Click to copy.
- **Actions** — Preview · Download · Send to clip editor.

> ☞ **SAVE TIME WITH SEARCH**
>
> Use the top-of-page search box to jump to a specific date + channel directly. Format is `{channel} {yyyy-mm-dd}`, e.g. `geo-news 2026-08-04`. The Recordings page will land you straight on that day's hour strip.

PART III · RECORDING & CLIPPING

09 Downloading segments

Pulling raw recording material off the platform for archival, evidence, or downstream editing.

One segment

The simplest case: you need the raw MP4 for one specific minute.

1. Drill down to the segment (Recordings → channel → day → hour).
2. Hit the **Download** action in that segment's row.
3. A signed URL is generated and the browser starts the download. Filename convention: `{channel}_{yyyy-mm-dd}_{HHMMSS}.mp4`.
4. Verify the download by comparing its SHA-256 to the one shown in the segment row.

A range of segments

Frequently you want a longer chunk — say, an hour of coverage. CMC will stitch adjacent segments server-side and hand you a single MP4.

1. Tick the checkbox on each segment you want, or drag the range across the segment list.
2. The floating *Bulk actions* bar appears at the bottom of the page.
3. Click **Download stitched**. The server concatenates the selected segments (they must be contiguous) and streams the merged MP4 as one download.

4 The stitched file's filename encodes the start and end times: `{channel}_{yyyy-mm-dd}_{HHMMSS}-{HHMMSS}.mp4`.

! **VIEWER ROLE RESTRICTION**

The Download and Bulk-download actions are hidden entirely for the `VIEWER` role. Viewers can preview segments but cannot move media off the platform. If a Viewer needs a copy, an Analyst or Admin has to do the download.

§ **EVIDENCE CHAIN**

Every download is written to the audit log with the timestamp, the operator, the segments involved, and the file's SHA-256. When a copy of a CMC recording is submitted as evidence, the audit log entry is how you prove chain-of-custody.

PART III · RECORDING & CLIPPING

Creating clips

10

The NLE editor — cut a precise moment out of a recording, brand it, export it.

A *clip* is a purpose-built cut of a recording, exported as a standalone MP4 for briefings, evidence submissions, or press replies. The clip editor is patterned after professional NLE (non-linear editor) tools — keyboard-driven, precise to the frame.

Opening the editor

There are three ways to open the editor with source material loaded:

- **From a segment row** — hit the "Send to clip editor" action. The editor opens with that segment on the timeline.
- **From a stitched selection** — pick a range of segments, hit **Create clip from selection**. The editor opens with that whole range on the timeline.
- **From scratch** — click **Clips** → **New clip** in the top nav. A dialog asks for the channel and time range; the editor opens on that material.

The editor layout

Two panels side by side:

- **Source** (left) — the raw material with its own playhead. This is what you scrub through to find the exact in and out points.
- **Program** (right) — the current cut with the watermark composited on. This is what the exported MP4 will look like.

Below both is the **timeline**: a horizontal ruler with a waveform showing the audio envelope, the current in/out markers as coloured bars, and a playhead you can drag with the mouse or step with the keyboard.

The clip workflow

1. **Find your in point.** Scrub the source until the playhead sits on the frame that should START the clip. Press `I`. The in-marker snaps to the playhead.

2. **Find your out point.** Scrub forward to the frame that should END the clip. Press `O`. The out-marker snaps to the playhead.

3. **Preview.** Press `Space` in the Program panel. Playback loops between the in and out markers. Adjust with `I` / `O` as many times as you like.

4. **Fill in the metadata.** The right sidebar asks for a title (required), notes (optional but recommended), and confirms your watermark. The watermark defaults to your CMC email + the current timestamp; the field is editable if you need a different tag.

5. **Pick format & resolution.** Dropdown lets you choose between MP4 at 1080p, 720p, or 480p. 720p is the sensible default — small enough to email, still crisp on a laptop screen.

6. **Click Export.** A toast confirms the export has queued and shows a progress bar. Depending on length the export takes 5–30 seconds; you can keep working elsewhere in CMC while it runs.

7. **The clip appears in Clips → Recent.** A toast link jumps you straight to it, ready to preview or download.

Transport shortcuts

SHORTCUT	ACTION
`Space`	Play / pause in the focused panel.
`J`	Reverse playback (repeat presses accelerate).

SHORTCUT	ACTION
`K`	Pause.
`L`	Forward playback (repeat presses accelerate).
`I`	Set in point at the current playhead.
`O`	Set out point at the current playhead.
`←` / `→`	Step one frame back / forward.
`Shift` + `←` / `→`	Step one second back / forward.

> ☞ **WATERMARK IS NON-OPTIONAL**
>
> Every exported clip carries a burned-in watermark in the bottom-right corner — your CMC email plus the export timestamp. This is deliberate: it makes casual re-sharing traceable. There is no way to disable it from the editor.

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PART III · RECORDING & CLIPPING

Managing clips

11

Finding, previewing, downloading, and (with permission) removing exported clips.

The clips list

The Clips page lists every clip on the platform, newest first. Each row shows the title, channel it came from, the covered time range, its duration, who exported it, and the export timestamp.

Filters along the top:

- **By channel** — narrow to one channel.
- **By operator** — narrow to clips exported by a specific person.
- **By date** — clips exported in a specific window.
- **Full-text search** — matches title AND notes.

Clip actions

ACTION	WHO CAN DO IT	WHAT IT DOES
Preview	VIEWER +	Opens an inline player. No download; strictly for review.
Download	ANALYST +	Downloads the MP4 to your machine. Logged in the audit trail.
Copy link	ANALYST +	Puts a signed, time-limited URL on your clipboard. Valid for one hour.
Delete	ADMIN	Removes the clip and its MP4 from disk. Cannot be undone.

Copy-link workflow — for evidence handover

Rather than emailing an MP4 attachment (which many mail relays strip), the *Copy link* action generates a short-lived URL that anyone with the link can use to download the clip.

1. Find the clip in the list.
2. Click **Copy link**. A toast confirms the URL is on your clipboard and shows its expiry.
3. Paste into your handover email. Recipient clicks it, browser downloads the MP4.
4. After the expiry (default one hour), the URL stops working. The clip itself remains on the platform.

> ! **DELETE IS IRRECOVERABLE**
>
> There is no trash. Once an Admin deletes a clip, both the row and the underlying MP4 are gone. If you need to keep a clip beyond retention, place a *legal hold* on the covering window (see Chapter 16) — that prevents both janitor deletion and manual delete.

PART IV · DOWNLOADS

12 Downloads panel

The single spot to fetch official CMC assets — installer, launcher, docs.

The **Downloads** page is a small central library of the assets operators need but that don't fit anywhere else in the app.

ASSET	WHAT IT IS	SIZE
CMC-VideoWall.zip	The desktop video-wall client, portable folder — no install needed.	≈ 45 MB
Software Manual (this document)	Latest revision of the operator manual, PDF.	≈ 3 MB

How a new build reaches the operator

When Analysys ships a revised CMC-VideoWall build it is uploaded to the Downloads folder on the server and becomes available to every operator immediately — nothing to re-run on the server.

1. The new ZIP replaces the previous one at `D:\cmc\downloads\CMC-VideoWall.zip`.
2. The Downloads page auto-serves whatever is at that path — the operator sees the fresh file on the next visit.
3. Operators download the new ZIP and extract it, replacing their local folder. Saved wall assignments and preferences carry across because those live in `%APPDATA%\CMC-VideoWall\`, not in the app folder.

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PART V · ADMINISTRATION

Users & roles

13

Creating operator accounts, controlling what each account can do, revoking access cleanly.

The user roster

Admin → Users. A table of every account, showing email, display name, role, last-sign-in timestamp, and status (active / disabled). Sortable and searchable.

Creating an account

1. Click **Add user**. A dialog opens.
2. Fill in email (must be unique platform-wide), display name (what the user will see in their menu), and an initial password.
3. Pick a role — ADMIN / OPERATOR / ANALYST / VIEWER. See Chapter 1 for the role model.
4. Click **Create**. The account is active immediately; hand the credentials over via a secure channel (voice, physically-handed print, encrypted chat) — not plain email.

Disabling access

When someone leaves the duty roster, *don't delete — disable*. Deleting removes the audit-log linkage; disabling keeps history intact while immediately terminating access.

1. Find the user in the roster.

2. Click the **Disable** icon in their row (a padlock).

3. Confirm the dialog. All active sessions for that account are killed within 60 seconds. The row moves to the "Disabled" section and greys out.

4. To re-enable later, click the padlock again.

Passwords & MFA

The current build uses email + password only. MFA is on the roadmap for a subsequent release. In the meantime:

- Passwords must be at least 12 characters with mixed case and a digit.
- Password re-use is not enforced by the platform — enforce operationally.
- An account is auto-locked after five consecutive failed sign-ins; Admin has to explicitly unlock.

> § **PASSWORD RESET**
> A forgotten-password self-service flow is deliberately absent — resets go through the Admin so there's an accountable human in the loop. Admin sets a new temporary password from the user detail page and hands it over out-of-band.

PART V · ADMINISTRATION

14 Audit log

Every state-changing action, immutable, forever — the accountability spine of the platform.

Every action in CMC that CHANGES state — a login, a source added, a clip exported, a segment downloaded, a user disabled — is written to the **audit log** as an immutable record. Reads (browsing recordings, previewing a clip) are not logged; only writes and access to media.

What each entry contains

- Timestamp (PKT, second-precision).
- Actor (the user who performed the action).
- Actor's role at the time (in case they've since been changed).
- Action verb (`SIGN_IN`, `CLIP_EXPORT`, `SEGMENT_DOWNLOAD`, …).
- Entity type + ID the action targeted.
- Diff — for updates, the before/after values.
- IP address of the client.

Browsing the log

Admin → Audit log. A reverse-chronological table. Filter down the volume quickly with:

- Time range picker — last hour, last day, last week, custom.
- Actor — pick one user or a group.
- Action verb — pick from a dropdown.
- Entity type — pick "Channel", "Clip", "User", "Segment", "Setting".
- Free-text search across all fields.

Exporting the log

Click **Export current view** to download the filtered result set as a CSV. Useful for compliance handovers or post-incident write-ups.

> ! **IMMUTABILITY**
>
> There is no interface anywhere in CMC to edit or delete an audit log entry. Entries can be exported and archived but never modified, which is what makes them useable as evidence. If you find a mistake in a log entry (unusual — most errors are actor mistakes at the time, not log corruption) file it as an incident, don't try to patch it.

PART V · ADMINISTRATION

15 Settings

The small set of platform-wide knobs an Admin actually needs to touch.

Admin → Settings. Every knob here is platform-wide and takes effect on save — no restart required.

SETTING	DEFAULT	EFFECT
Clip default resolution	720p	What the clip editor's resolution dropdown shows as pre-selected.
Clip URL expiry	1 hour	How long a *Copy link* URL stays valid.
Session timeout	8 hours	How long a signed-in browser session lasts before requiring a re-login.
Air-gap mode	Off	When on, blocks every outbound connection from the frontend (fonts, images, telemetry). For deployments with no internet.
Password min length	12	Minimum length enforced on account creation and password change.

§ **EVERY CHANGE IS LOGGED**

Changing any of these values is a state change — every save writes an audit log entry showing the before/after values and who made the change.

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PART V · ADMINISTRATION

16 Legal holds

Keeping specific windows of recording safe from janitor deletion — indefinitely.

The 30-day retention is fine for most material. But when an incident is under review, a specific time window needs to be kept regardless of how much time passes. That's a **legal hold**.

A legal hold names a channel (or "all channels"), a start and end time, a reason, and the person who requested it. While it's active, the janitor will REFUSE to delete any segment that falls inside that window, even if it's older than the retention days.

Placing a hold

1. Admin → Legal holds.
2. Click **New hold**.
3. Pick a channel (or "all"), enter start and end timestamps, write a reason. The reason field is not optional — this is what an auditor will read six months from now.
4. Click **Place hold**. The hold is active immediately and shown in the roster with a padlock icon on the affected channel row.

Expiring / releasing a hold

Holds either expire naturally at their end time OR are released manually. Releasing a hold does not automatically delete the covered segments — it just makes them eligible for deletion at the next janitor run if they are also outside the normal retention window.

! **HOLDS WIN OVER DELETE**

A hold blocks both the janitor AND manual clip deletes across the covered window. If you attempt to delete a clip whose material falls inside an active hold, the operation is refused with a toast pointing at the responsible hold. This is intentional: it's the accountability rail.

PART VI · REFERENCE

17 Troubleshooting

The symptoms you actually see, the first action that fixes them 90% of the time.

Web app symptoms

SYMPTOM	LIKELY CAUSE	FIRST ACTION
Sign-in returns "Wrong email or password"	Bad credentials, or account disabled.	Ask Admin to verify the account exists and is active.
Sign-in returns "Session expired"	Browser session timed out.	Retry — it should sign in cleanly.
Dashboard KPIs stuck / not updating	WebSocket dropped, page lost live feed.	Reload the page (F5).
Channel signal red for one channel	Satellite feed dropped, or source URL wrong.	Open the channel, check the sources list; alert IT if all sources red.
Recording calendar shows no days	Wrong channel picked, or channel not recording.	Confirm the channel is enabled for recording in Admin → Channels.
Clip export fails with "cannot stitch non-contiguous segments"	The picked segments have a gap between them.	Choose smaller ranges each side of the gap, export separately.

Video Wall symptoms

SYMPTOM	LIKELY CAUSE	FIRST ACTION
App won't launch — VCRUNTIME140.dll missing	Older ZIP without Visual C++ runtime bundled.	Re-download the latest ZIP from Downloads; extract fresh.
App launches but window never appears	OpenGL missing — usually on RDP without hardware GPU.	Run at the physical console instead of RDP, or ensure opengl32.dll is present next to the exe.
"Device is not ready" on cmd launch	Extracted to an unmountable path (removable drive ejected).	Re-extract to a local drive (C:\ or D:\).
Some tiles black on the wall	Those channels' UDP sources aren't reaching the wall PC.	Verify the Multicast NIC IP is correct at login; verify with VLC that `udp://@239.1.1.5:1111` plays.
Grid-32 tiles stutter	CPU saturated — normally 32 concurrent decodes on the operations wall PC. On a laptop, use Mosaic (20) instead.	Switch to Mosaic-20 layout.
ESC doesn't return from fullscreen	Focus is not on the wall window.	Click the fullscreen window first, then press Esc.

☞ **WHEN THE FIX ISN'T OBVIOUS**

Every symptom above assumes you're seeing SOMETHING. If the app is utterly unresponsive, capture the console log to a file — `CMC-VideoWall.cmd > log.txt 2>&1` — and send it to Analysys support. The log almost always names the failing subsystem in its last few lines.

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PART VI · REFERENCE

A Appendices

Shortcut sheet, URLs, and how to reach us.

A · Full keyboard shortcut sheet

CLIP EDITOR

SHORTCUT	ACTION
`Space`	Play / pause.
`J` / `K` / `L`	Reverse / pause / forward transport.
`I` / `O`	Set in / out marker.
`←` / `→`	Step one frame back / forward.
`Shift` + `←` / `→`	Step one second back / forward.
`Ctrl` + `Enter`	Export clip.

VIDEO WALL (FULLSCREEN MODE)

SHORTCUT	ACTION
`F11`	Toggle fullscreen (enter or exit).
`Esc`	Exit fullscreen back to preview.
Right-click on tile	Fill mode & clear-slot menu.

B · URL reference

PURPOSE	URL
Web app (recording LAN)	`http://10.51.80.202:3001`
Web app (storage LAN)	`http://172.16.205.21:3001`
API (used by Video Wall)	`http://10.51.80.202:4001`
Downloads folder (server)	`D:\cmc\downloads\`
Recording data (server)	`/data/cmc/recordings/`

C · Support & escalation

For issues with the CMC software itself that survive the first-action fixes in Chapter 17:

Analysys support desk

Primary contact for software bugs, feature requests, and incident-response on the platform. Email and phone details supplied in the deployment handover pack.

Escalation window

P1 (platform unusable): 30 minutes to acknowledgement, 4 hours to resolution. P2 (feature not working as expected): next business day.

End of manual

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